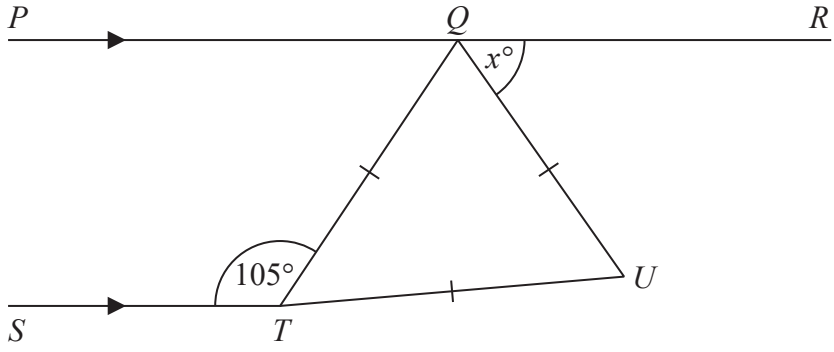
17



PQR is a straight line parallel to *ST*.
QUT is an equilateral triangle.

Angle *STQ* = 105°

Work out the value of *x*.
Give a reason for each stage of your working.

Angle TQU = 60°: equilateral triangle.

1 mark
Equilateral angle 60°.

Angle PQT = 180° – 105° = 75°:

1 mark
Find 75°.

co-interior angles on parallel lines add to 180°.

75° + 60° + x = 180°: angles on a straight line.

1 mark
Use the straight line.

x = 180° – 75° – 60° = 45°.

1 mark
45° with matching reasons.

(Total for Question 17 is 4 marks)